



Unequal Plates: Drivers of Racial Disparities in Food Insecurity Across the U.S.

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MATH445

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Introduction:

Background and Motivation:

Food insecurity is an issue that has been and continues to be relevant every single day. To be food insecure is to have a lack of consistent access to food. This doesn't mean that someone has to be worrying about food daily, rather that they could have to worry about not having enough to eat in even just a day of the week sometimes. This issue affects millions of people in the U.S. every year.

As with many things, racial and ethnic disparities exist within food insecurity. Black and Hispanic populations experience a rate of food insecurity higher than White (non-hispanic) populations. Understanding possible underlying causes in these disparities is essential to developing and fixing the gap and bettering food security as a whole. And as recent policy changes (the Big Beautiful Bill, which modified SNAP eligibility) further restrict access to food assistance, looking deeper into the biggest factors of racial disparities in this subject is increasingly prevalent.

Research Questions:

For this report and project, our overarching question/motivation is:

What factors contribute the most to racial disparities in food insecurity?

Furthermore, we developed the following sub-questions to help us answer that:

1. Is it just that states with high overall FI have bigger gaps?
2. Do economic factors (meal cost) matter more than policy factors (SNAP)?
3. Are the main reasons for the Black-White gap the same as the Hispanic-White gap?

Answering these questions and understanding the relationship between variables can help to inform possible policy interventions and reduce any gaps in racial disparities.

Data Overview:

Data Description:

The data used in this report comes from a dataset called *Map the Meal Gap*, a dataset produced by Feeding America. This dataset provides annual county and state-level estimates of food insecurity across the United States, spanning 2019-2023. The dataset provides us with a wide range of indicators relating to food insecurity, including economic conditions, SNAP information, and demographics. Some of the most important variables we paid most attention to include:

- Food Insecurity Measures
 - Overall food insecurity rate
 - Food insecurity among racial/ethnic demographics
- Economic Indicators
 - Average cost per meal
 - SNAP Threshold

Data Sources:

The *Map the Meal Gap* dataset is produced by Feeding America in collaboration with researchers and public data sources. In order to produce this data, Feeding America uses a combination of national survey data, public government statistics, as well as statistical modeling to estimate food insecurity rates. Key data sources include:

- Current Population Survey (CPS)
- American Community Survey (ACS)
- Bureau of Labor Statistics (BLS)

These data sources allowed Feeding America to generate consistent, comparable estimates of food insecurity across all U.S. counties and states.

Data Pre-Processing:

Data Cleaning:

To clean the data, we took three core steps.

1. Renamed Columns
 - Column names weren't easy to work with, so we made standardized names for everything. For example, any mention of "food insecurity" became "fi", "%" became "pct", and we utilized underscores instead of spaces.
2. Mutated Columns
 - Many of the columns were numbers that had symbols within their values (";", "%", "\$"). We parsed the numbers out of these values to make them usable in our visualizations and regression.
3. Added Columns
 - To answer our research questions, we calculated the difference between the food insecurity rates for black and white people, as well as hispanic and white people. We then used the mutate function to add this as a new column for analysis.

None

```
set.seed(123)
food <- food %>%
  rename(
    fips = FIPS, state = `State Name`,
    state_abbr = State, year = Year,
    fi_rate_overall = `Overall Food Insecurity Rate`,
    fi_num_overall = `# of Food Insecure Persons Overall`,
    black-fi_rate = `Food Insecurity Rate among Black Persons (all ethnicities)`,
    hispanic-fi_rate = `Food Insecurity Rate among Hispanic Persons (any race)`,
    nonhispanic-fi_rate = `Food Insecurity Rate among White, non-Hispanic
Persons`,
    snap_threshold = `SNAP Threshold in State`,
    pct-fi_below_snap = `% FI ≤ SNAP Threshold`,
    pct-fi_above_snap = `% FI > SNAP Threshold`,
    child-fi_rate = `Child Food Insecurity Rate`,
    fi_children = `# of Food Insecure Children`,
    pct-fi_child_below_185fpl = `% food insecure Children in HH w/HH Incomes Below 185
FPL`,
    pct-fi_child_above_185fpl = `% food insecure Children in HH w/HH Incomes Above
185 FPL`,
    fi_rate_senior = `Senior Food Insecurity Rate`,
```

```

fi_seniors = `# of Food Insecure Seniors`,
fi_rate_older_adult = `Older Adult Food Insecurity Rate`,
fi_older_adults = `# of Food Insecure Older Adults`,
cost_per_meal = `Cost Per Meal`,
weekly_budget = `Weighted weekly $ needed by FI`,
annual_budget_shortfall = `Weighted Annual Food Budget Shortfall`,
census_division = `Census Division`,
census_region = `Census Region`,
fns_region = `FNS Region`
) %>% mutate(across(c(fi_rate_overall,
                      black_fi_rate,
                      hispanic_fi_rate,
                      nonhispanic_fi_rate,
                      snap_threshold,
                      child_fi_rate,
                      fi_rate_senior,
                      fi_rate_older_adult,
                      pct_fi_below_snap,
                      pct_fi_above_snap,
                      pct_fi_child_below_185fpl,
                      pct_fi_child_above_185fpl,
                      cost_per_meal,
                      weekly_budget,
                      annual_budget_shortfall), ~ parse_number(.x)))

```

None

```

food_gaps <- food %>%
  mutate(
    bw_gap = black_fi_rate - nonhispanic_fi_rate,
    hw_gap = hispanic_fi_rate - nonhispanic_fi_rate,
    bw_ratio = black_fi_rate / nonhispanic_fi_rate,
    hw_ratio = hispanic_fi_rate / nonhispanic_fi_rate) %>%
  select(
    state, year, census_region, census_division, bw_gap, hw_gap,
    bw_ratio, hw_ratio, fi_rate_overall, cost_per_meal, snap_threshold,
    black_fi_rate, hispanic_fi_rate, nonhispanic_fi_rate
  )

```

Exploratory Data Analysis:

Before modeling, we examined potential trends and regional patterns in the data to understand how food insecurity varies across racial and ethnic groups over time.

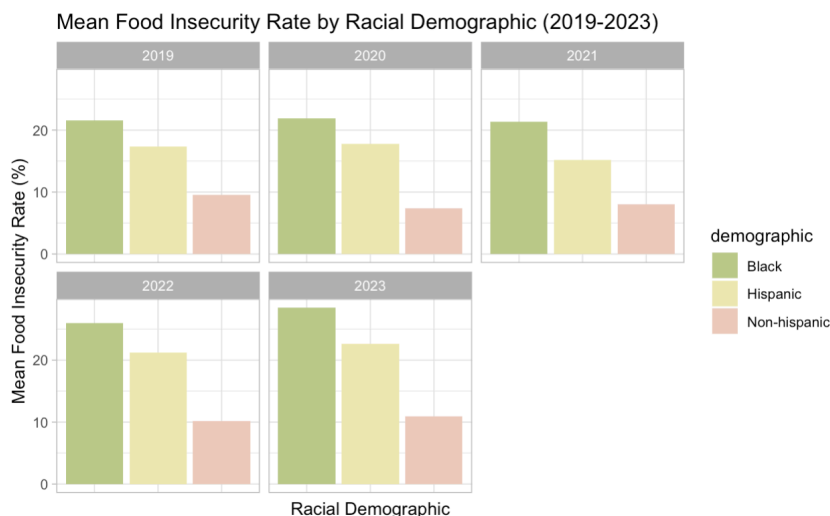


Figure 1: Mean food insecurity rates by racial/ethnic group, 2019–2023

- Clear disparities exist, with Black and Hispanic populations consistently experiencing higher food insecurity rates than White populations across all five years.
- Black populations have the highest food insecurity, peaking at around 28% in 2023.

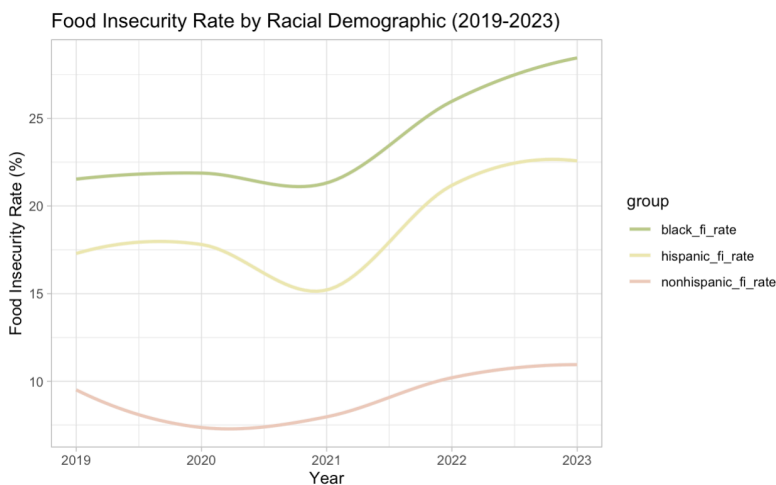


Figure 2: Mean food insecurity rates by racial/ethnic group over time

- Food insecurity rates have increased over time, but it is clear the rate of growth differs substantially across racial/ethnic groups.
- We can observe a noticeable dip in Black and Hispanic food insecurity rates.
 - This reflects increased government assistance programs during the COVID-19 pandemic, including stimulus checks and enhanced SNAP benefits.

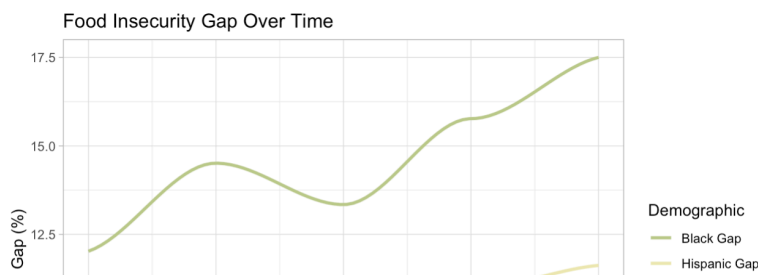


Figure 3: Black–White and Hispanic–White food insecurity gaps over time

- This graph shows how Black and Hispanic food insecurity gaps have changed over time, with a noticeable dip in 2021, followed by a continued increase in disparities.

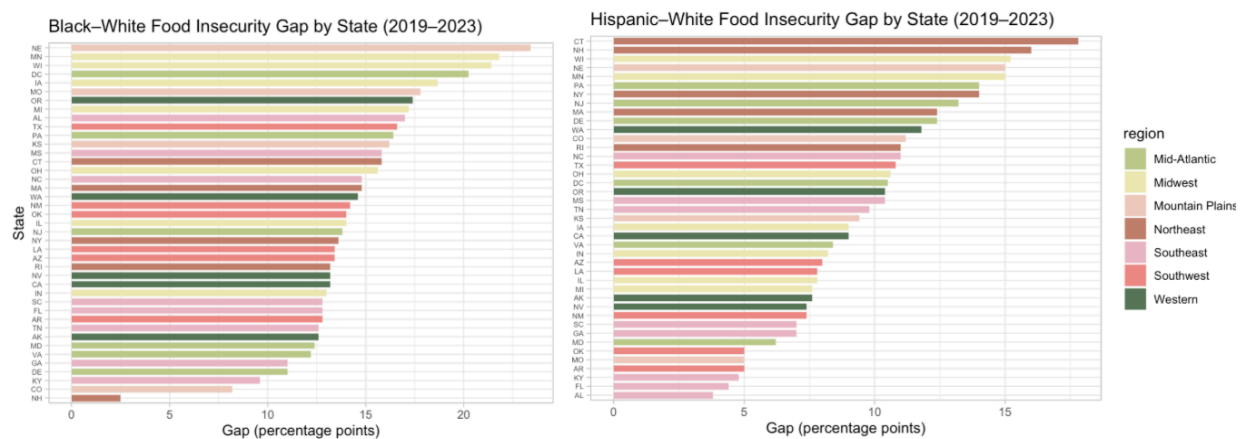


Figure 4: Black–White and Hispanic–White food insecurity gaps by State and Region

- Minnesota, Nebraska, and Wisconsin display elevated gaps for both populations.
- Nebraska shows the largest Black-White food insecurity gap.
- Connecticut shows the largest Hispanic-White food insecurity gap.

The graphs reveal a persistent trend in racial and ethnic disparities in food insecurity, with Black and Hispanic populations consistently experiencing higher rates than White populations. A temporary decline in food insecurity occurred during the COVID-19 pandemic, likely due to government aid programs, however, disparities quickly resumed once these programs were halted. While certain states display elevated gaps, a proper conclusion cannot be made regarding regional data with the lack of population data. Overall, these observations highlight the ongoing gaps and disparities of food insecurity across racial and ethnic groups.

Methodology Analysis:

Statistical Approach:

We employed the use of multiple linear regression modeling, taking on the form:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_p x_p + \epsilon, \epsilon \sim N(0, \sigma_\epsilon)$$

- y is the outcome variable (Black-White Gap, Hispanic-White Gap)
- x_i are predictor variables (Meal Cost, SNAP threshold, etc)
- β_j are the coefficients that are being estimated by the model
- ϵ is the random error accounted for

Bootstrap Sampling:

To make sure our data does not rely on normality assumptions (as we do not know if our data is normally distributed), we used bootstrap resampling over 300 iterations. The bootstrapping process we used was as follows:

1. Randomly samples (with replacement) observations from the original dataset
2. Fits the regression model on each bootstrap sample
3. Collects all coefficient estimates across all bootstrap samples
4. Calculates the standard deviation of the bootstrap distribution, as well as the 95% confidence intervals

We fit two models, one for the black-white disparity and one for the hispanic-white disparity. In our model, although we are more interested in economic and policy factors, controlling census regions was integral for making sure we saw more accurate results. Unobserved regional differences (historical, cultural, or structural) within the regions are then controlled for. This ensures that the coefficients represent effects within regions to increase accuracy.

None

```
bw_gap_model <- lm(bw_gap ~ fi_rate_overall +  
                  cost_per_meal + snap_threshold +  
                  census_region, data = food_gaps)  
bw_boot_gaps <- food_gaps %>%  
  bootstraps(times = 300) %>%  
  mutate(  
    model = map(splits, ~lm(bw_gap ~ fi_rate_overall +  
                           cost_per_meal + snap_threshold +  
                           census_region, data = analysis(.x))),  
    coefs = map(model, tidy)  
  ) %>%
```

```

unnest(coefs)

bw_boot_gaps %>%
  filter(term %in% c("fi_rate_overall", "cost_per_meal", "snap_threshold")) %>%
  group_by(term) %>%
  summarize(
    mean_estimate = mean(estimate),
    bootstrap_se = sd(estimate),
    ci_lower = quantile(estimate, 0.025),
    ci_upper = quantile(estimate, 0.975)
  )

```

```

None

hw_gap_model <- lm(hw_gap ~ fi_rate_overall +
                  cost_per_meal + snap_threshold +
                  census_region, data = food_gaps)

hw_boot_gaps <- food_gaps %>%
  bootstraps(times = 300) %>%
  mutate(
    model = map(splits, ~lm(hw_gap ~ fi_rate_overall +
                           cost_per_meal + snap_threshold +
                           census_region, data = analysis(.x))),
    coefs = map(model, tidy)
  ) %>%
  unnest(coefs)
summary(hw_gap_model)

hw_boot_gaps %>%
  filter(term %in% c("fi_rate_overall", "cost_per_meal", "snap_threshold")) %>%
  group_by(term) %>%
  summarize(
    mean_estimate = mean(estimate),
    bootstrap_se = sd(estimate),
    ci_lower = quantile(estimate, 0.025),
    ci_upper = quantile(estimate, 0.975))

```

Results:

Black-White FI Gap:

Variable	Estimate	Std. Err.	t-val.	p-val.	Boot Estimate	Boot Std. Err	Lower CI	Upper CI
Overall Food Insecurity Rate	0.490	0.091	5.372	2.19e-07	0.493	0.091	0.322	0.674
Cost per Meal	3.072	0.555	5.537	9.77e-08	3.08	0.579	1.97	4.17
SNAP Threshold	0.021	0.008	2.664	0.008	0.021	0.007	0.008	0.036

Figure 5: Regression results and bootstrapped coefficient estimates for predictors of Black-White food insecurity gaps.

Model $R^2 = 0.4217$, Adjusted $R^2 = 0.4041$

Key Findings:

Cost Per Meal seems to be the strongest predictor. Each \$1 increase in the cost per meal is associated with a 3.08% increase in the gap. This is the most significant predictor in the model, having the smallest p-value. This level of effect signifies that as grocery prices and food costs increase, Black people face a greater amount of food insecurity compared to white people.

Overall Food Insecurity Rate also matters. Each 1% increase in the overall rate correlates with a 0.49% increase in the gap. This suggests that when food insecurity increases, it does not affect every racial group the same way, and that black populations have to deal with amplified impacts.

Unexpectedly, the SNAP threshold has a positive relationship with the gap, leading to a 0.021% increase. The SNAP threshold being higher *should* intuitively lead to less food insecurity, as more people fall into the income limit and thus hypothetically qualify. However, more SNAP eligibility is associated with larger racial gaps, which can reflect several things:

- Unequal Access: Even when eligible, black households may face greater barriers to SNAP enrollment due to documentation requirements, lack of information, and the complexity of the administrative process.
- Structural Barriers: Lack of access to transportation, work schedule conflicts, or just stigma surrounding the topic may prevent black households from accessing SNAP benefits.

Model Fit:

The model R^2 (0.4217) indicates that around 42% of the variation in Black-White food insecurity gaps across states can be explained by our predictors. This may not seem like a lot on the surface, but in the realm of social science research, human behavior is influenced by plentiful unmeasured factors. Thus, a 0.4 is considered reasonably strong depending on the circumstances.

Bootstrap Validation:

All of the Bootstrap estimates for coefficients are close to the actual estimates. The same goes for the respective standard errors. As well, in our 95% confidence interval for all three variables, the values within them do not contain zero or fall into the negatives, affirming statistical significance besides from the p-values.

Hispanic-White FI Gap:

Variable	Estimate	Std. Err.	t-val.	p-val.	Boot Estimate	Boot Std. Err	Lower CI	Upper CI
Overall Food Insecurity Rate	0.173	0.087	1.992	0.047	0.168	0.083	-0.002	0.320
Cost per Meal	1.918	0.520	3.689	0.0003	1.89	0.462	0.992	2.79
SNAP Threshold	0.034	0.007	4.618	6.50e-06	0.034	0.007	0.02	0.05

Figure 6: Regression results and bootstrapped coefficient estimates for predictors of Hispanic-White food insecurity gaps.

Model $R^2 = 0.4086$, Adjusted $R^2 = 0.3929$

Key Findings and Comparisons:

Cost Per Meal remains a significant predictor in the Hispanic-White gap as it was in the Black-White gap. Each \$1 increase in the cost per meal is associated with a 1.918% increase in the gap. This is around a percent less of a gap, meaning that economic factors affect black

populations a lot more. However it is still indicative of a disparity in how grocery prices affect Hispanic households.

Overall Food Insecurity Rate is less significant here. Each 1% increase in the overall rate correlates with a 0.173% increase in the gap. It also has the greatest p-value, just barely falling under 0.05, and the confidence interval we bootstrapped includes 0, which displays less significance than is the case for the Black-White gap.

The SNAP threshold has a stronger relationship here, having a much smaller p-value. Every 1% increase in SNAP threshold leads to a 0.034% increase in the gap. This is much larger than the case for the Black-White gap, showing that Hispanic populations are more likely to be affected by things like language barriers and immigration status concerns.

Model Fit:

The model R^2 (0.4086) indicates that around 41% of the variation in Hispanic-White food insecurity gaps across states can be explained by our predictors. This is a similar fit as the Black-White gap, and as discussed before, relevant because this is within the realm of social science.

Bootstrap Validation:

All of the Bootstrap estimates for coefficients are close to the actual estimates. The same goes for the respective standard errors. However, in the 95% confidence interval for Overall FI rate, it includes 0, which indicates a weaker significance for that variable. Otherwise, the other two seem very significant based on their p-values and the confidence intervals.

Discussion:

Answering our Research Questions:

1. Is it just that states with high overall FI have bigger gaps?
 - Overall food insecurity rate is a significant predictor of the Black-White gap, but not so significant for the Hispanic-White gap. This indicates that underlying dynamics and other factors are at play here.
2. Do economic factors (meal cost) matter more than policy factors (SNAP)?
 - Higher meal costs are strongly associated with bigger gaps for both groups than the SNAP threshold. Meaning, lowering the cost of meals will reduce the gap in food insecurity for minority groups. Increasing the SNAP threshold indicates that the gap would increase, but we have to look at this more carefully.
 - The positive coefficient of the SNAP threshold relationship likely indicates that states with larger gaps respond to this by increasing the generosity of the threshold. However, this potential policy response does not address the underlying issues and the possible barriers to SNAP programs that prevent eligible minority households from enrolling.
3. Are the main reasons for the Black-White gap the same as the Hispanic-White gap?
 - Generally, for the variables we looked into, yes. However there are important differences in magnitude.
 - Meal Costs affect both gaps significantly, but for the Black-White gap, it has a 3% gap, which is 50% more than the Hispanic-White gap of 1.9%.
 - Overall FI rate strongly predicts the gap for Black-White, but it is seemingly insignificant for the Hispanic-White.
 - SNAP Threshold effects are closer in magnitude for both, but larger for the Hispanic-White gap (0.034% vs 0.021% for Black-White).
 - The differences suggest that although both the black and hispanic populations of the US face disparities in food insecurity, Black people are more affected by the conditions of the economy whereas Hispanic people are influenced more potentially by barriers to immigrant communities.

Policy Implications:

Our findings suggest several policy priorities that can possibly be enacted to reduce racial disparities in food insecurity:

- Reducing or removing taxes on food items
 - Many states have reduced or eliminated taxes on food items in grocery stores, but some still have them, which can be contributing to this disparity.
- Vouchers for fresh produce
 - Fresh produce tends to be more expensive, so by distributing vouchers to marginalized communities, it can reduce costs and promote healthier eating.
- A simplified, reinforced approach to the SNAP application
 - Making it easier to apply for SNAP means that it becomes more accessible to everyone, and more people would qualify and benefit.
 - Ensuring that immigrants don't feel threatened by applying is crucial. As well, making sure that it is accessible for those who don't speak English.
 - Increasing outreach to encourage more people to apply and decrease the stigma for applying.

Limitations:

Modeling Limitations:

All values in the dataset are modeled estimates derived from survey and economic data, rather than direct counts. The way these values are calculated may introduce a degree of uncertainty, as we cannot count on their complete accuracy. These estimates rely on statistical methods that may smooth over local variation, meaning they should be interpreted with caution, especially at the county-level.

Precision Limitations:

Demographic group data may be less precise or reliable in areas with smaller Black or Hispanic populations. This limitation primarily affects county-level data, but this may still occur among state-level data. In these smaller populations, sample sizes may be limited to produce precise estimates.

Contextual Limitations:

This dataset does not provide population data among demographic groups. The lack of population data greatly restricts the ability to properly assess racial/ethnic disparity among different areas. Additionally, several local economic factors and conditions that could influence food insecurity and its estimates are not provided, including:

- Cost of living
 - Housing and utility
 - Transportation
 - Healthcare
 - Childcare
- Employment conditions
 - Local unemployment rates
 - Underemployment
 - Availability of living-wage jobs

These missing contextual factors make it difficult to fully understand food insecurity and to accurately assess the true extent of racial and ethnic disparities within.

Future Directions:

To address the limitations identified in the dataset, as well explore other crucial dimensions of this large issue, there are several directions for future work on this project.

- Incorporating missing contextual factors
 - We can use another dataset that includes crucial population and demographic data, allowing us to look more closely at regional disparities rates and better interpret what these differences truly represent.
- Analyze at the county-level
 - Examining county-level data to capture localized variation can give us a better understanding of how disparities manifest across different areas.
- Revisit the current dataset
 - Due to the recent *One, Big, Beautiful Bill Act*, critical changes are being made to SNAP eligibility and funding.
 - It's crucial to come back to this dataset in the coming years, allowing us to assess how these policy changes are reflected in food insecurity trends and racial disparities.

Addressing these limitations and the stated future directions will provide us with the necessary tools to draw stronger, more meaningful conclusions relating to racial disparities in food insecurity.

Conclusion:

This analysis highlights persistent racial disparities in food insecurity, with Black and Hispanic populations consistently experiencing higher rates than White populations. Among the factors involved, cost per meal displayed the strongest relationship with disparity gaps, increasing around 2-3% every \$1 increase.

SNAP eligibility threshold and overall food insecurity rate had notable impacts as well, but this varied greatly between the two demographic groups examined. In order to address these disparities, policy interventions that reduce food costs and improve access to food assistance could help tighten the gap. However, the data had several limitations.

Limitations include modeled estimates and missing population data, which suggest interpreting some results with caution. In the future, we want to address these limitations by incorporating additional demographic and population data, examining county-level patterns, and revisiting the data as recent policy changes take effect. These efforts will allow for stronger conclusions and contribute to a more equitable future.

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